PHYSICS

Ultrafast spin transfer and its impact on the electronic structure

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Optically induced intersite spin transfer (OISTR) promises manipulation of spin systems within the ultimate time limit of laser excitation. Following its prediction, signatures of ultrafast spin transfer between oppositely aligned spin sublattices have been observed in magnetic alloys and multilayers. However, it is known neither from theory nor from experiment whether the band structure immediately follows the ultrafast change in spin polarization or whether the exchange split bands remain rigid. We show that ultrafast spin transfer occurs even in ferromagnetic gadolinium metal. Charge transfer between localized surface and extended valence-band states leads to a decrease of the surface spin polarization. This synchronously alters the exchange splitting of the bulk valence bands during laser excitation. Moreover, the onset of demagnetization can be tuned by over 200 fs by changing the temperature-dependent spin mixing. Our results show a promising route to ultrafast control of the magnetization, widening the impact and applicability of OISTR.

INTRODUCTION

Since the first observation of femtosecond demagnetization (*1*) and subsequently of all-optical magnetic switching (*2*), the field of ultrafast magnetism has been driven by the questions of how fast and by which microscopic process spins and magnetic order can be manipulated by light (*3*). These problems not only are of fundamental importance but also have obvious technical implications for data manipulation and ultrafast spintronic applications (*4*, *5*). In first order, light couples only to electronic but not to spin excitations. Therefore, ultrafast manipulation of the spin polarization requires transfer of electronic to spin excitations by means of exchange interaction (*6*–*8*), spin-orbit coupling (*9*), spin-lattice coupling (*10*), and/or transport of spins by hot electron (*11*) or magnon currents (*12*). At interfaces and in alloys, exchange scattering opens an additional channel to transfer angular momentum between differently aligned spin sublattices (*13*). All these processes require coupling between spin, electron, and/or phonon subsystems of a solid, and their response will depend on scattering rates and relaxation times.

In 2018, Dewhurst *et al.* (*14*) predicted that already laser excitation can induce spin-selective charge flow, which changes the magnetic structure of materials, including switching from antiferromagnetic to ferromagnetic order. Since this spin modulation is purely optical, it belongs to the fastest processes manipulating magnetization by light. Follow-up experimental studies using x-ray magnetic circular dichroism (XMCD) and the transversal magneto-optical Kerr effect revealed first indications of optically induced intersite spin transfer (OISTR) attributed to light field–driven coherent charge relocation (*15*, *16*). More recent studies related the faster and more efficient demagnetization of a CoPt alloy as compared to pure Co metal to OISTR (*17*) and revealed a transient ferromagnetic alignment of antiferromagnetic Mn in Co/Mn multilayers (*18*). Very recent transient magneto-optical Kerr effect (T-MOKE) studies observed intersite spin transfer in a Heusler half-metal (*19*) and ultrafast spin transfer effects in FeNi alloys (*20*).

These experiments support the concept of ultrafast intersite spin transfer predicted by time-dependent density functional theory. However, the impact of OISTR on the electronic band structure and, in particular, its response time are known neither from theory nor from experiment. Does the optical transfer of spin polarization lead to an immediate change in exchange splitting faster than the coupling to the lattice discussed in (*21*–*23*) and potentially even faster than ultrafast spin excitations reported in (*24*–*25*)?

In the present article, we reveal signatures of optically induced spin transfer in the valence electronic structure of gadolinium metal. Here, optical excitation leads to spin transfer between localized surface and delocalized bulk states, a novel channel of OISTR. Time-resolved photoemission reveals a depletion of the majority spin surface state and of the minority spin bulk bands. This constitutes a spin-selective charge flow between surface and bulk. It manifests in a transient shift of the majority spin surface state toward lower binding energy and a simultaneous increase of the bulk-band exchange splitting. The timescale of spin transfer is set by the pump-pulse duration. The concomitant response of the valence bands appears on the same timescale within experimental accuracy. Since Gd shows spin mixing, we can increase the spin polarization in the valence bands by decreasing the initial sample temperature. This allows us to enhance OISTR and postpone the onset of ultrafast bulk demagnetization by about 200 fs. Our results directly prove the impact of OISTR on the magnetization dynamics and its control by spin mixing.

Among the experimental probes of magnetization dynamics, time- and angle-resolved photoemission spectroscopy (tr-ARPES) with vacuum ultraviolet (VUV) radiation is a rather demanding but specific probe of the magnetic system. It directly measures population

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dynamics in the time domain (*26*) and provides momentum-selective information on electronic surface states as well as bulk bands up to a few-monolayer escape depth, set by the inelastic mean free path of the photoelectrons. This surface sensitivity allowed us to follow the spin flow and exchange splitting at the Gd(0001) surface, which hosts a localized $d_z{}^2$-derived surface state. Its majority spin component $d^{\uparrow}_{z^2}$ is occupied, and its minority spin counterpart was found above the Fermi level E_F (*27*, *28*). The exchange splitting of bulk bands and surface state mainly reflects the magnetization of the 5*d* moments (*22*). As illustrated in Fig. 1A for $T \rightarrow 0$ K, the Gd valence bands are exchange split and fully spin polarized (*29*, *30*). In the majority spin channel (blue), a near infrared (NIR) pump pulse excites electrons from the occupied $d^{\uparrow}_{z^2}$ surface state into the unoccupied bulk bands (*28*). In the minority spin channel (red), optical excitation lifts electrons from the occupied $(5d6s)^3$ bulk bands into the unoccupied $d^{\downarrow}_{z^2}$ surface state. In consequence, the spin polarization decreases at the surface and increases in the near-surface bulk layers. This particular scenario will be true for most rare-earth metal surfaces (*31*) and can be generalized to magnetic multilayers with spin-polarized interface states. For elevated temperatures, the exchange splitting decreases and surface and bulk bands become spin mixed (*27*, *30*). Spin mixing was unambiguously demonstrated by scanning tunneling spectroscopy of the occupied and unoccupied Gd surface state (*27*). While the exchange splitting between $d^{\uparrow}_{z^2}$ and $d^{\downarrow}_{z^2}$ remains finite at the Curie temperature ($T_C = 297\ K$) assuming about 60% of its value at liquid He temperature (*27*), the spin polarization vanishes at T_C (*32*). This complete band mirroring but finite exchange splitting at T_C was likewise observed for the Gd Δ_2-like Σ bulk bands in spin-resolved ARPES (*30*). As illustrated in Fig. 1B, when approaching the Curie temperature T_C, band mirroring by spin mixing makes the optical transitions decreasingly spin sensitive; identical transitions occur in both spin channels and overall diminish the effect of OISTR.

RESULTS

To follow charge and concomitant spin transfer at the Gd surface, we performed tr-ARPES measurements with *p*-polarized probe

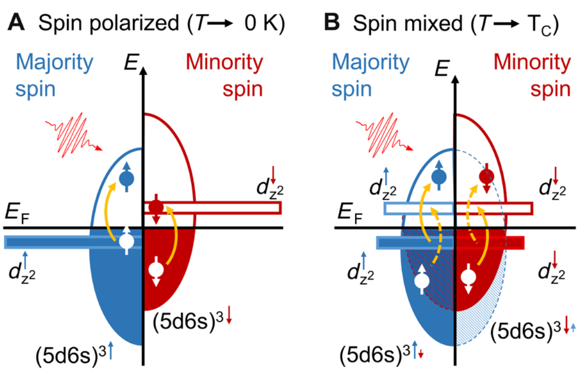

Fig. 1. Optically induced spin transfer. OISTR at the Gd(0001) surface occurs between bulk-like $(5d6s)^3$ valence bands and occupied majority ($d^{\uparrow}_{z^2}$) and unoccupied minority ($d^{\downarrow}_{z^2}$) spin components of the Gd surface state. (**A**) For temperatures $T \rightarrow 0$, states are highly spin-polarized and optical excitation leads to spin transfer from surface to bulk states. (**B**) When *T* approaches the Curie temperature T_C, spin mixing leads to band mirroring and diminishes OISTR.

pulses stemming from higher-order harmonic generation (HHG). We used two different setups, denoted as RAL and FUB, both based on Ti:sapphire laser amplifiers, as described in Materials and Methods.

Figure 2 shows time-resolved photoemission measurements in a delay range of −1 ps to +2 ps of the VUV probe to the NIR pump pulses. The energy scale on the ordinate refers to the Fermi level E_F of the Gd metal film. Individual electron-distribution curves (EDCs; see the Supplementary Materials) were recorded with the RAL setup at 3° off-normal with an angular resolution of ±3° and stitched to a false color map. The IR pump and VUV probe pulses had a photon energy of 0.95 and 34.2 eV, respectively. Sample temperature was 50 K.

A striking feature of the time-resolved data is the oscillations in kinetic energy E_{kin} at negative delay. As already analyzed by Bovensiepen *et al.* (*33*), these oscillations arise through ponderomotive acceleration of the emitted photoelectron in the transient grating formed by interference of the delayed incident and reflected *s*-polarized pump pulse in front of the metal surface. From the oscillation amplitude, we can reasonably well extract the absorbed pump fluence to be 1.3 ± 0.3 mJ/cm^2. We extended the simulations in (*33*, *34*) to verify that we can safely ignore modulations of E_{kin} at positive delays. On this basis, we hereinafter analyze spin transfer and valence-band dynamics following laser excitation.

At the chosen VUV photon energy, we map the Gd majority spin surface state $d^{\uparrow}_{z^2}$ near the center of the surface Brillouin zone (SBZ) $\overline{\Gamma}$, and the exchange-split Δ_2-like Σ bulk bands ($d^{\downarrow}$ and $d^{\uparrow}$) in the fourth bulk Brillouin zone near Γ (*35*).

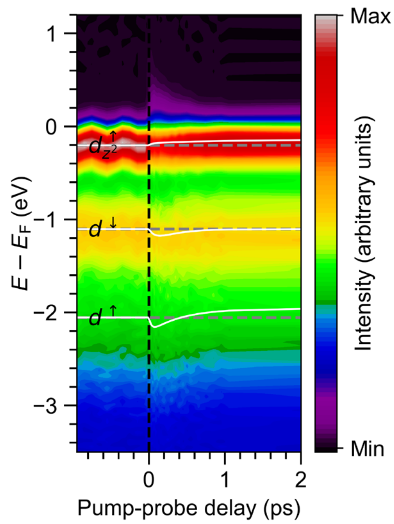

Fig. 2. Electron dynamics and transient band structure near the Γ point of Gd(0001). Photon energies of pump and probe pulses were 0.95 and 34.2 eV, respectively. The NIR pump pulses were *s*-polarized, and the VUV probe pulses were *p*-polarized. The dashed vertical line marks zero delay. The white lines indicate the peak positions and are the fitting results of Fig. 4. Optical excitation leads to a depletion of the majority spin surface-state $d^{\uparrow}_{z^2}$ as well as an increase of hot electrons above the Fermi level E_F. Upon laser excitation, $d^{\uparrow}_{z^2}$ shifts upward to lower binding energy, while the minority and majortity spin bulk bands ($d^{\downarrow}$ and $d^{\uparrow}$) shift initially downward to higher binding energy. The oscillations in binding energy at negative pump-probe delay are of no magnetic origin (see Results).

OISTR between surface and bulk states

It is immediately obvious from the tr-ARPES raw data in Fig. 2 that there is a sizeable decrease in intensity of the occupied component of the $d^{\uparrow}_{z^2}$ surface state (red) building up around zero pump-probe delay (dashed vertical line) accompanied by an increase of the electron population above E_F (violet).

The corresponding dynamics of the hot electron distribution above the Fermi level and of the population of the majority spin surface state are summarized in Fig. 3 (see fig. S1). For the moderate absorbed pump fluence of ~1.3 mJ/cm^2 we observe a ~22% decrease of the $d^{\uparrow}_{z^2}$ peak area and a corresponding increase of the photoemission intensity above the Fermi level. The solid lines are fits to the data, both modeled by an exponential function convolved with an error function of 42 ± 10 fs width. This time constant reflects the cross-correlation of pump and probe pulses, i.e., the temporal resolution of the RAL experiment. Decay of the hot electron population above E_F occurs during the equilibration of electron and phonon subsystems and can be described by a single exponential with a time constant of 260 ± 30 fs (*36*). In contrast, the recovery of the surface state population shows a slower time constant of 400 ± 80 fs. Since the NIR pump pulse depletes the surface-state band, electrons at the SBZ center $\overline{\Gamma}$ probed in the photoemission experiment will relax to higher binding energies along the downward dispersing surface-state band (*35*). This additional decay channel explains the delay in population recovery at $\overline{\Gamma}$.

Ultrafast response of the band structure synchronous to OISTR

The binding energy of the valence states changes synchronously with the population dynamics. The solid white lines in Fig. 2 highlight the peak positions. They correspond to the fitting results from Fig. 4. The $d^{\uparrow}_{z^2}$ surface state shifts upward to lower binding energy. The OISTR effect is best seen in the EDCs, depicted in fig. S1. During optical excitation, we observe a clear depopulation of the surface state and a concomitant up-shift by about 40 meV to lower binding energy. In addition, we observe a downward shift of the minority spin $d_{\downarrow}$ bulk band (yellow) to higher binding energies within ~150 fs after laser excitation. Finally, we note that the majority spin $d^{\uparrow}$ bulk band (green) shows in the raw data a small protrusion of its high-energy rim (at ~3.05 eV) for positive delays up to ~200 fs, indicating a transient downward shift to higher binding energy.

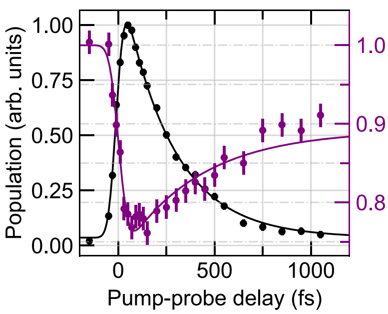

Fig. 3. Population dynamics. Buildup and decay of the hot electron population above the Fermi level (integrated intensity in the range of $E - E_F = 0.5$ to 1.0 eV, black circles) as well as depletion and population recovery of the occupied majority spin surface state $d^{\uparrow}_{z^2}$ at $\overline{\Gamma}$ (purple circles, intensity of peak maximum). The solid lines are fits to the data (see "OISTR between surface and bulk states" section and fig. S1).

To extract the transient band positions, we decomposed the EDCs in individual peaks comprising the occupied majority spin component of the Gd surface-state as well as the exchange-split minority and majority spin valance d bands (see fig. S2) (*35*). A complete set of globally fitted EDCs as a function of pump-probe delay may be found in fig. S3. Figure 4 depicts the transient binding energies of the $d^{\uparrow}_{z^2}$ surface state and the minority $5d^{\downarrow}$ and majority $5d^{\uparrow}$ spin components of the valence band as well as the exchange splitting $\Delta E_{ex} = E_B(5d^{\uparrow}) - E_B(5d^{\downarrow})$ of the bulk bands. We use double-exponential fits (solid lines) to qualitatively describe the dynamics (see the Supplementary Materials, section E). Time constants τ are listed in Table 1.

The signature of OISTR in the electronic structure directly manifests in the ultrafast response of both majority spin states, $d^{\uparrow}_{z^2}$ and $5d^{\uparrow}$. Peak positions of surface and bulk state shift upon optical excitation within the cross-correlation of pump and probe pulses (gray shaded area in Fig. 4). The upward shift of $d^{\uparrow}_{z^2}$ to lower binding energy results in a depletion of the majority spin surface-state

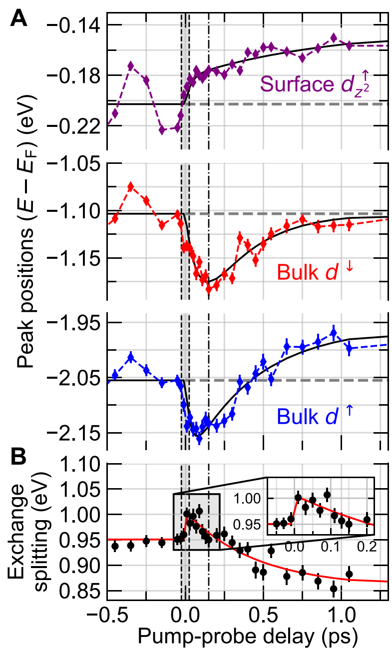

Fig. 4. Valence-band dynamics. (**A**) Binding energy of the $5d^{\uparrow}_{z^2}$ surface state and the minority and majority spin bulk bands ($5d^{\downarrow}$ and $5d^{\uparrow}$) as a function of pump-probe delay. Error bars show two standard deviations. The dashed horizontal lines are the average of the binding energies at negative pump-probe delay. The solid lines are double-exponential fits to the data. Time constants τ_1 and τ_2 are listed in Table 1. For more details, see the Supplementary Materials, section E. During optical excitation (cross-correlation indicated by gray area), the binding energy of the majority spin surface state decreases, while that of the majority spin bulk band increases. The opposite shift of the majority spin bands reflects the pump pulse–induced spin transfer from surface to bulk (see Fig. 1). In line, the $5d^{\downarrow}_{z^2}$ minority spin valence band shows a slowed down response reaching maximal binding energy not until 150-fs pump-probe delay (dashed-dotted vertical line). (**B**) Transient exchange spitting Δ_{ex} of the bulk bands. We observe a small but significant increase of Δ_{ex} within the first 100 fs after optical excitation (error bars show two standard deviations; see also fig. S4). With a time constant of 370 ± 50 fs, electron and phonon subsystems equilibrate (see fig. S6). On a comparable time scale the binding energies of the valence bands shift leading to a ~10% reduced bulk exchange splitting which characterizes the transiently demagnetized state.

Table 1. Time constants (in femtoseconds) of double-exponential fits to the transient binding energies in Fig. 4.

	τ_1	τ_2
$d_{z^2}^{\uparrow}$ surface state*	7 ± 15	740 ± 430
$5d^{\downarrow}$ bulk band	82 ± 45	290 ± 160
$5d^{\uparrow}$ bulk band	34 ± 12	450 ± 75

*τ_1 and τ_2 are strongly correlated, leading to large error bars, but a single exponential describes the data significantly worse (see fig. S5).

population (see Fig. 3 and fig. S1) and a concomitant decrease of the spin polarization at the surface.

A surprising finding is the initial downward shift of the majority spin valence band $5d^{\uparrow}$ to higher binding energy. For ultrafast demagnetization, we would expect a reduction of the exchange splitting and thus an upward shift of the majority spin bulk band, which is only observed at larger delays ≥150 fs. Instead, we observe the signature of an increase of the bulk magnetization, which we attribute to OISTR from surface to bulk ($d_{z^2}^{\uparrow}$ surface state to $d^{\uparrow}$ bulk band).

Like the majority spin bulk band $5d^{\uparrow}$, the minority spin bulk band $5d^{\downarrow}$ shifts first to higher binding energies. However, $5d^{\uparrow}$ and $5d^{\downarrow}$ do not react in parallel. When we compare the minimum band positions (maximum binding energies) of $5d^{\uparrow}$ and $5d^{\downarrow}$ in Fig. 4, we see that they are reached at clearly different delays of about 50 and 150 fs (see also fig. S5). Accordingly, we obtain distinct time constants τ_1 of 34 and 84 fs describing the fast dynamics (Table 1). We will discuss this complex behavior in "Tuning ultrafast spin transfer via spin mixing."

Finally, on the slower timescale of electron-phonon equilibration ($\tau \sim 370$ fs; fig. S6), all three bands move to lower binding energies with time constants τ_2 of a few hundred femtoseconds (see Table 1).

DISCUSSION

Transient increase of the bulk-band exchange splitting

The observed pump-induced shift of the valence bands could be in part of electronic and not of magnetic origin, e.g., due to localization and band narrowing (*37*). However, such a shift has not been observed in time-resolved XMCD of Gd (*38*). Moreover, it is very unlikely that we observe opposite electronic shifts in the manifold of occupied bands, for the valence *d*-band and the *d*-band derived surface state. We conclude that the dynamics of the exchange splitting Δ_{ex} are of purely magnetic nature. The dynamics of the bulk exchange splitting are depicted in Fig. 4B.

We find a small but significant increase of Δ_{ex} by ~40 meV relative to the equilibrium value before laser excitation (inset of Fig. 4B). The rise of Δ_{ex} parallels the increase of the electron temperature T_e. It follows the laser excitation and is thus the fingerprint of OISTR.

Unfortunately, the tr-ARPES data (Fig. 2) do not allow us to follow the unoccupied minority spin component of the surface state. From a 1.5-eV pump and 6-eV probe photoemission experiment, Lisowski and colleagues (*28*, *39*) concluded that the exchange splitting of the surface state remains constant but only within experimental error of 100 meV. Furthermore, the depletion of the occupied majority spin component of the surface state leads to a reduction of the overall surface spin polarization. This was confirmed by the ultrafast drop of the magnetic second harmonic generation (SHG) signal (*40*). In turn, majority bulk states above E_F become occupied. This leads to an increase of the spin polarization of the bulk bands during optical excitation.

As long as dipole transitions are considered, the optical transfer of spin polarization between surface and bulk remains local and limited to the near surface, i.e., within the coherence length of the bulk states. Because of the characteristic minimum of the electron inelastic mean-free-path *l* in metals, photoemission with VUV photon energies is surface sensitive. Our tr-ARPES experiment probes the near-surface region ($l \sim 3$ to 5 layers, ≤10 Å), and we can observe the small but significant transient increase of the bulk exchange splitting. For a rough estimate of the change in exchange splitting, we assume that the 23% depopulation of the surface state leads to a comparable increase of the bulk majority spin states distributed over five layers. The majority spin bulk density of states (DOS) at E_F is about one state per electron volt and two atoms (i.e., number of atoms per unit cell). Here, we used the very similar, calculated bulk DOS of Tb from figure S8 in (*41*). If we assume a DOS of 0.5 states per atom for the majority spin surface state, we would obtain a value of 44 meV for the shift of the exchange splitting, which is in the right ballpark ($\Delta E_{ex}[\mathrm{eV}] \cdot 5/2 = 0.5 \cdot 0.23$).

At approximately 100-fs delay, Δ_{ex} falls to its start value. Already during optical excitation, the electronic system starts to thermalize via inelastic electron-electron scattering. Typical electron quasiparticle lifetimes at 1 eV above the Fermi level, e.g., in the 3*d* metals, are 10 fs (*42*, *43*). Via direct or exchange scattering, the decay of a photoexcited electron or hole creates an electron-hole pair in the same or opposite spin channel (*6*, *44*). When exchange scattering dominates, as, e.g., shown for cobalt (*44*), the optically induced spin polarization of the bulk states will diminish after few scattering events. This explains why the transient increase of the bulk exchange splitting lasts only for about 100 fs. In addition, carriers optically excited into bulk states may escape via ballistic transport into deeper layers of the bulk (*11*). Free electrons with a kinetic energy of 1 eV travel 5.9 Å/fs. If net, ballistic spin transport from the surface into the bulk dominated, we would expect an even weaker and shorter transient increase of the bulk exchange splitting.

A time-resolved MOKE study on Gd was performed at similar pump fluence (*45*). Sultan *et al.* (*45*) observed a slight, transient increase in MOKE rotation upon laser excitation of Gd at 40 K sample temperature. However, since this effect was not found in the MOKE ellipticity, it was classified as an optical artifact and corrected. From a later study on cobalt, it became obvious that MOKE rotation and ellipticity probe different depth profiles of the sample (*46*). The MOKE rotation is more surface sensitive than the ellipticity. This suggests that the transient increase of the MOKE rotation in Gd is not an optical artifact but corroborates our findings and reflects the initial increase of the bulk magnetic moment and exchange splitting due to OISTR.

After the transient increase, the exchange splitting drops to ~90% of its initial value with a time constant of 450 ± 75 fs. This mainly reflects the dynamics of the majority spin bulk band $d^{\uparrow}$, which shows a larger shift than $d^{\downarrow}$ (see Fig. 4 and Table 1). The drop of the electron temperature T_e, i.e., the equilibration of electron and lattice subsystems, occurs on a similar timescale of 370 ± 50 fs (fig. S6). This suggests that following the initial buildup of the bulk spin polarization and its decay via electron-electron exchange scattering, electron-phonon spin-flip scattering leads to demagnetization of the valence bands. The dynamics on even longer timescales (not discussed here) is set by 4*f* spin to phonon coupling with a time constant of about 15 ps (*22*, *32*, *41*). We note that the decrease of the exchange splitting due to Elliott-Yafet spin flip scattering showed no counterpart in the 4*f* dynamics (*22*) and did not alter the spin polarization of the surface state (*32*). However, changing the spin polarization of the bulk bands (via OISTR or spin currents) may affect the 4*f* spin system. We found a small initial drop of the 4*f* magnetic moment in (*22*), which was, however, within two standard deviations.

Tuning ultrafast spin transfer via spin mixing

For the here studied pump fluences, optical excitations are dominated by dipole transitions between the occupied and unoccupied bands. Thus, changing the band structure, e.g., by varying the start temperature, should directly influence optically induced charge and/or spin transfer. As already illustrated in Fig. 1, this is particularly true for Gd metal, since in thermal equilibrium it is a prototypical example for spin mixing: With increasing sample temperature, majority and minority spin bands become more and more mirrored (see Fig. 1). Thus, changing the sample temperature allows us to tune the spin polarization of the initial and final states in optical excitation. Figure 5 (A and B) compares transient binding energies and exchange splitting for three initial sample temperatures of 60, 100, and 200 K. The corresponding spin polarization of valence bands and surface state amount to about 80, 70, and 55%, respectively (*27*, *30*, *32*). The temperature-dependent time-resolved photoemission measurements were performed using the FUB setup with 1.6-eV pump pulses and 36.8-eV HHG probe pulses. The absorbed pump fluence was 3.5 ± 1 mJ/cm^2.

Let us first examine the decay of the exchange splitting. Fitting a single exponential (solid line in Fig. 5B), we obtain time constants τ of 840, 860, and 870 fs, which match within error bars (see Fig. 5B). As already seen in the RAL experiment (see Fig. 4 and fig. S6), for the different settings used in the FUB experiment, τ compares again well with the time constant of ~800 fs characterizing the decay of the electron temperature T_e (insets in Fig. 5B). This corroborates our finding that equilibration between optically excited electron and initially cold phonon subsystems drives the decay of the 5*d* exchange splitting via Elliott-Yafet spin-flip scattering. We note that Δ_{ex} is proportional to the 5*d* magnetic momentum (*22*).

The overall drop of the exchange splitting $\delta\Delta_{ex}$ becomes smaller with increasing temperature (Fig. 5B). Therefore, the 5*d* demagnetization rate $\propto\delta\Delta_{ex}/\tau$ in units of meV/fs changes from about 250/840 = 0.3 at 60 K to 190/860 = 0.22 at 100 K and 170/870 = 0.2 at 200 K. The net spin-flip scattering rate and thus the demagnetization rate

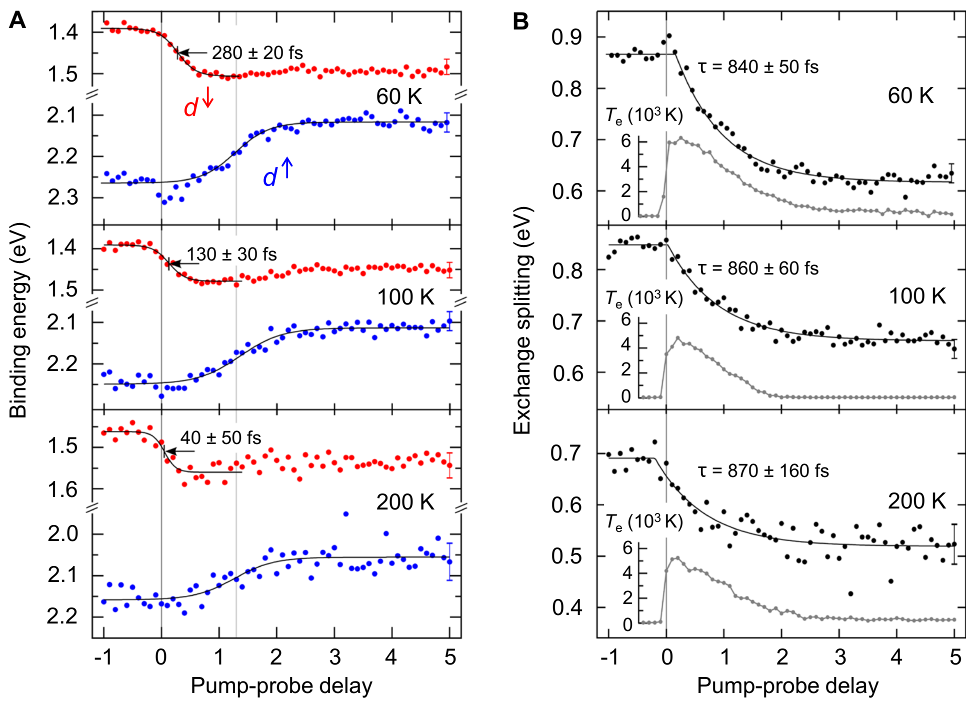

Fig. 5. Transient electronic structure of Gd for different sample temperatures and spin mixing. (**A**) Binding energies of the minority (red) and majority (blue) spin valence-band components as a function of pump-probe delay. (**B**) Corresponding evolution of the exchange splitting. The single exponential fits (solid lines) yield similar decay constants for all three temperatures, but the exchange splitting changes by 250 ± 10, 190 ± 12, and 170 ± 24 meV for 60, 100, and 200 K, respectively. Temporal overlap of pump and probe pulses (i.e., absolute delay zero) was verified by the jump in the electron temperature (gray). Error bars are, for the sake of clarity, only included for the last data point and show two standard deviations.

should scale with the difference of (excited) majority and minority spin electrons $n^{\uparrow} - n^{\downarrow}$, which is proportional to the overall spin polarization $(n^{\uparrow} - n^{\downarrow})/(n^{\uparrow} + n^{\downarrow})$. Changing the spin polarization from 0.55 at 200 K to 0.7 at 100 K and 0.8 at 60 K (*27*, *30*), we expect an increase in demagnetization rate $\delta\Delta_{ex}/\tau$ by a factor of $0.8/0.55 = 1.45$ and $0.7/0.55 = 1.27$, which is in the ballpark of the measured values of 1.5 ± 0.4 and 1.3 ± 0.6. Differences may be easily explained by the slightly different pump fluences, judged from the maximum electron temperature T_e (inset in Fig. 5B) as well as by uncertainties in determining the sample temperature and in extrapolating the spin polarization.

We now turn to the dynamics around zero delay. To avoid space charge in the FUB experiment, the 770-nm pump pulses needed to be stretched to a pulse duration of about 250 fs (*47*). This smears out the initial drop in the band dynamics seen in the RAL experiment (see Fig. 4). In the FUB experiment (Fig. 5), we find at 60 K sample temperature only few data points outside two standard deviations indicating the dip in the $d^{\uparrow}$ band position and the transient increase of the exchange splitting around zero delay. However, we clearly observe a shift of the drop of the minority spin bulk band $d^{\downarrow}$ to earlier delays by $280 - 130 = 150$ fs and by $280 - 40 = 240$ fs, when comparing the sample temperature of 60 K with 100 K and 200 K, respectively. This shift translates into the drop of the exchange splitting starting at similarly earlier delays. For all start temperatures, delay zero aligns with respect to the initial increase of the electron temperature T_e as shown in the insets of Fig. 5B. This guarantees that the shift of minority spin band and exchange splitting with start temperature are genuine. As already discussed, the optical excitation leads to an effective spin transfer from surface to bulk states. This results in an increase of the bulk exchange splitting and delays the onset of its decay. For higher sample temperature, spin mixing increases and the effect of optically induced spin transfer diminishes. Accordingly, we observe a shift of the minority spin band and of the onset of demagnetization to earlier delay times.

Let us now look back at the RAL experiment, which was recorded at 50 K sample temperature. In this experiment, OISTR must be notable due to the high spin polarization of the valence bands. Upon optical excitation, both $5d^{\downarrow}$ and $5d^{\uparrow}$ bulk bands shift to higher binding energies but with different time constants of 82 and 34 fs, respectively. It is this combination of the shift of the spin majority bulk band to higher binding energy and the delayed response of the minority spin bulk band that leads to a temporary increase of the exchange splitting during optical excitation. As this is not expected for optically driven demagnetization via coupling between spin, electron, and/or phonon subsystems, we argue that both effects are signatures of OISTR.

In a T-MOKE study of permalloy, Mathias *et al.* (*48*) observed a shift of ~18 fs between the demagnetization of iron and nickel state selectively probed at the $M_{2,3}$ edges. Very recently optically induced spin transfer from the occupied Ni minority spin channel into the Fe minority spin channel was theoretically predicted and experimentally observed for a $Ni_{50}Fe_{50}$ alloy (*16*, *20*). Combining these results with our observation of distinct band shifts, we conclude that the delayed response of Ni with respect to the Fe demagnetization reflects the different shifts of bands with predominant Ni or Fe character. Please note that in our experiment on Gd, it is the distinct localized and extended charge distribution in surface and bulk states that allows us to observe the signature of ultrafast spin transfer in the electronic structure. One may argue that a ~40-meV increase of the exchange spitting is just a tiny 4% effect. While this is true, please note that in contrast to theory (*14*) our RAL experiment was limited to a rather low pump fluence of 1.3 mJ/cm^2 to avoid space-charge effects in time-resolved photoemission with high temporal resolution.

In conclusion, we have used femtosecond time-resolved ARPES to observe signatures of OISTR in the exchange-split band structure of an elemental ferromagnet. During laser-driven demagnetization, the electron population, spin polarization, and exchange splitting show comparable ultrafast dynamics. Thus, OISTR not only changes the spin polarization on an ultrafast timescale but also affects synchronously the position of valence bands and exchange splitting. OISTR may therefore have the potential to coherently control not only the transient spin polarization but also truly the magnetization dynamics. This may be used for an ultrafast terahertz gate, temporarily changing the magnetization at the interfaces of a multilayer stack. In addition, we have demonstrated that OISTR is sensitive to spin mixing and, hence, can be tuned by the sample temperature.

MATERIALS AND METHODS

Photoemission measurements have been performed in two independent experimental apparatuses. The setup in Berlin (FUB) delivers VUV pulses at 10-kHz repetition rate (*47*). The laser fundamental ($\lambda = 770$ nm, $h\nu = 1.61$ eV) was used as *s*-polarized pump pulse. Photoelectrons were detected after a hemispherical electron analyzer. Because of the low work function of Gd ($\leq$3.7 eV), the pump pulses had to be stretched to avoid space charge, which limited the time resolution of the experiment to 250 fs at a pump fluence of $F \sim 3.5$ mJ/cm^2. Experiments with better time resolution of 42 ± 10 fs were performed at the Artemis experiment of the Central Laser Facility at the Rutherford Appleton Laboratory (RAL) (*49*). The 1-kHz setup uses a low-dispersion monochromator with off-plane mount gratings, giving an HHG pulse duration at the sample of 30 fs. In addition, we used pump pulses generated by optical parametric amplification with lower photon energy of 0.95 eV ($\lambda = 1300$ nm). Since absorption of four pump photons is required to overcome the Gd work function, pump pulse–induced space-charge effects were sufficiently suppressed. This allowed us to apply pump pulses of below 30-fs duration at an absorbed pump fluence of $F = 1.3$ mJ/cm^2. Photoemission spectra were recorded with a time-of-flight spectrometer (*50*). With a photon energy of 34.2 eV (RAL setup) and 36.8 eV (FUB setup) in tr-ARPES, we probe the third bulk Brillouin zone of the Gd(0001) hcp lattice in the $\Gamma - M$ direction. This reveals the minority and majority spin components of the Δ_2-like Σ valence band (*51*), denoted as $d^{\downarrow}$ and $d^{\uparrow}$, respectively (*22*). In both experimental apparatuses, epitaxial Gd films of ~10-nm thickness were grown on a W(110) substrate using identical homebuilt evaporators. The deposition rate was 7 Å per minute at pressures below 8×10^{-10} mbar. After Gd deposition at 300 K, films were annealed to 720 K for 60 s. Surface order was controlled by low-energy electron diffraction.

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S6
Table S1

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